4. TF-IDF [Term-frequency - Inverse Document frequency].

$S_1 \rightarrow$ good boy

$S_2 \rightarrow$ good girl

$S_3 \rightarrow$ boy girl good

$$\text{Term freq (TF)} = \frac{\text{No. of rep of word in sentence}}{\text{No. of words in sentence.}}$$

$$\text{IDF} = \log_e\left(\frac{\text{No. of sentences}}{\text{No. of sentences containing the word}}\right).$$

Term Frequency * **IDF** — BOW

	S_1	S_2	S_3
good	1/2	1/2	1/3
boy	1/2	0	1/3
girl	0	1/2	1/3

good	$\log_e(3/3) = 0$
boy	$\log_e(3/2) =$
girl	$\log_e(3/2)$

1	1	0	
1	0	1	
1	1	1	

Final TF-IDF — O/P

	good	boy	girl
Sent 1	0	$1/2 * \log_e(3/2)$	0
Sent 2	0	0	$1/2 \log_e(3/2)$
Sent 3	0	$1/3 \log_e(\frac{3}{2})$	$1/3 \log_e(3/2)$

- **Advantage and Disadvantage.**

Advantage

i. Intuitive

ii. Fixed Size → Vocab size

iii. Word Importance is getting captured.

Disadvantage

i. Sparsity still exists

ii. OOV.